

Tutorial-5
ME-101, (Engineering Mechanics, Semester-II)
Date: 19/02/2024 and Time: 7.55 to 8.50 AM

1. A person of mass $m_1 = 80$ kg kneeling on a plywood panel of mass $m_2 = 10$ kg wishes to push himself and the panel away from the right-hand wall as shown in **Figure 1**. Determine a condition on μ_2 which would allow him to do this, as opposed to his slipping relative to a stationary panel. The two applicable coefficients of static friction are $\mu_1 = 0.60$ and μ_2 .

[10 mark]

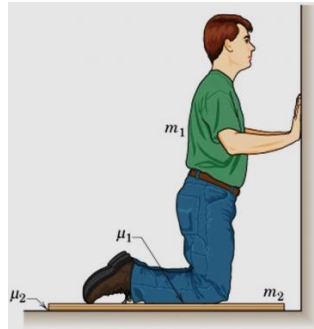


Figure 1

2. The vertical position of 100 kg block is adjusted by the screw activated wedge as shown in **Figure 2**. Calculate the moment M which must be applied to the handle of the screw to raise the block. The single threaded screw has square threads with mean diameter of 30 mm and advances 10 mm for each complete turn. The coefficient of friction for screw thread is 0.25 and all mating surfaces of the block and wedge is 0.4. Neglect friction at the ball joint A.

[10 mark]

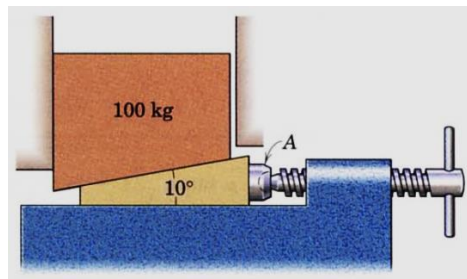


Figure 2

3. The tape slides around the two fixed pegs of diameter 30 mm as shown in **Figure 3** and is under the action of the horizontal tensions $T_1 = 40$ N and $T_2 = 160$ N. Determine the coefficient of friction μ between the tape and the pegs.

[10 mark]

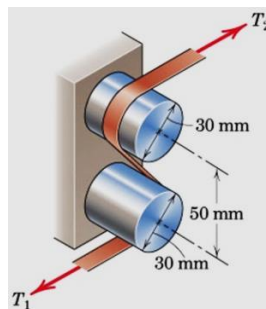


Figure 3

4. The system of two blocks, cable, and fixed pulley is initially at rest as shown in **Figure 4**. The horizontal force P necessary to cause motion when (a) P is applied to the 10-kg block and (b) P is applied to the 20-kg block. **[10 marks]**

- Draw free body diagram for case a **[1 mark]**
- Calculate the normal reaction forces between the blocks and between the lower block and fixed surface. **[2 marks]**
- Determine force P on 10 kg block and tension T in the cable for case a **[3marks]**
- Draw free body diagram for case b **[1 mark]**
- Determine force P on 20 kg block and Tension T in the cable for case b **[3 marks]**

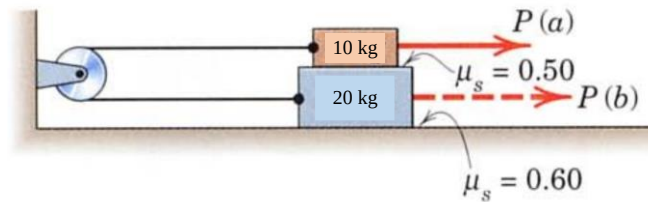


Figure 4

5. **Figure 5** shows a 5-kg uniform serving table being supported on each side by pairs of two identical links, AB and CD , and springs CE . The bowl has a mass of 1 kg. The springs each have a stiffness of $k = 200$ N/m and are unstretched when $\theta = 90^\circ$. Neglect the mass of the links. **[10 marks]**

- Draw the Free Body Diagram. **[2 marks]**
- Write the spring force expression in terms of angle θ . **[1 mark]**
- Write the virtual displacement expressions for the loads. **[3 marks]**
- Write the virtual work equation. **[1 mark]**
- Determine the angle θ where the table is in equilibrium. **[2.5 marks]**
- Determine the spring force when the table is in equilibrium. **[0.5 marks]**

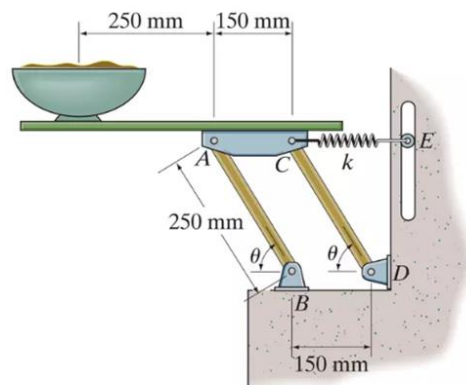


Figure 5